Pharmaceutical Formulation 2

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmaceutical Formulation 2 (A09661)

Short Title: Pharmaceutical Formulation 2
Faculty Science and Computing
Department Science
Cluster Pharmaceutical Science
Author ADONEGAN
Credits 10 **Level:** Intermediate

Description of Module / Aims

This module will give students a good understanding of the development on a range of solid, liquid and topical preparations for human and veterinary use. An introduction to the formulation and testing of sustained release products will also be covered. The importance of the device and the formulation in nasal inhalation products will be addressed in this module.

Programmes

PHAR-0004	BSc in Good Manufacturing Practice and Technology (WD_SGMPT_D)	1 / 3 / M

Indicative Content

- Emulsion formulation and manufacture; the use of surfactants, individual and combinations thereof and preservative systems in stabilising an emulsion
- Gels: excipients used in gel preparations; bioadhesion & formulation of bioadhesives
- Formulation of sustained release dosage forms; encapsulation and tabletting of slow release granules, matrix tablets
- Production of veterinary products; EC guidelines, MRL's in foods; setting of withholding periods
- Production of nasal sprays & aerosols; testing and quality control requirements
- Specifications for the control of dosage forms studied at release and during shelf life

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Propose formulations for the development of emulsion preparations.
2. Formulate gel preparations choosing appropriate excipients to achieve a suitable product.
3. Interpret the critical points to be considered in the production and testing of veterinary pharmaceuticals.
4. Plan the steps to be undertaken in formulating a sustained release product.
5. Classify the component parts and importance of same for Nasal/MDI preparations.
6. Formulate and prepare a solid oral dosage pharmaceutical preparation and carry out physical tests thereon. Prepare a suspension preparation.
7. Compile appropriate specifications for the control of the quality of a pharmaceutical product both at release and during shelf life.

Learning and Teaching Methods

- Lectures.
- Practicals.
- Industrial visits.
- E-learning via Moodle.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,3,4,5
Continuous Assessment	50%	
In-Class Assessment	25%	1,2,7
Lab Report	25%	6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Lab		12
Independent Learning		234

Supplementary Material

- "EU Legislation ." ec.europa.eu/health/documents/eudralex
- Lachman, L. _The theory and Practice of industrial pharmacy_. 3rd ed. Philadelphia: Lea and Febiger, 1986.
- Rowe, R.C. _Handbook of Pharmaceutical Excipients_. 7th ed. London: Pharmaceutical Press, 2012.

Requested Resources

- Room Type: Chemistry Lab